



KABARAK UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND IT

A WEB-BASED INTERN MANAGEMENT SYSTEM (IMS)

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A project/ proposal submitted in partial fulfillment for the requirement of the award of
Bachelor of Science in Computer Science Degree of Kabarak University.

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DECLARATION

I hereby declare that this research proposal is my original work and has not been submitted to any other institution for academic credit or assessment. All sources of information used in this work have been duly acknowledged and referenced in accordance with the required academic standards.

I further affirm that this proposal has been prepared independently and reflects my own understanding of the research topic.

Name of Student:

Registration Number:

Signature:

Date:

RECOMMENDATION

This project proposal has been submitted for examination with my approval as the university supervisor.

Name: Dr. Evans Kimutai

Designation: Supervisor

Signature:.....

Date:

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ABSTRACT

Internship programs are a cornerstone to tertiary education by bridging the gap between academic theory and professional practice. However, the digitised formal management of interns is absent in majority of small to medium business economies. The document delves into an information system solution, an Intern Management System, integratable into enterprise environments within the context of Kenyan organisations.

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CHAPTER ONE

INTRODUCTION

1.1 Introduction

This chapter serves as the foundation of the web-based Intern Management System (IMS), outlining its background and context, the problem it addresses, and its purpose. It begins with a the background of study which covers the general overview of the Intern Management System (IMS) which sheds light on how organisations currently rely on ad hoc solutions to manage intern processes, and the operational gaps this creates. Next, through the problem statement, it defines the specific problem the project aims to solve. Following, are the main and specific objectives, and research questions governing the study follow. In addition, the chapter proposes the core modules of the application, their functions, and contribution to the solution. The significance of the study validates the significance of the project while the scope of study curbs potential scope creep.

Together, these elements establish the foundation upon which the system is built.

1.2 Background of the Study

Internships and industrial attachment programmes are a cornerstone to tertiary education systems, serving as the primary mechanism through which students transition from academic theory to professional practice.

Organisations regard interns as a contingent workforce aiding in sourcing early talent. Said programmes aid both parties in attaining their goals: organisations source early talent from their contingent workforce while interns gain practical exposure while fulfilling their academic requirements.

According to the National Association of Colleges and Employers (NACE, 2023), majority of large organisations in large organisations in developed economies operate structured internship programmes. The digital platforms supporting functions such as applicant tracking, and performance evaluation, were widely adopted in North America and European organisational contexts.

In East African context, Kenya specifically, the Universities (Amendment) Act (2012) and the Technical and Vocational Education and Training (TVET) Act (2013), mandate industrial attachment as a prerequisite for degree or diploma completion.

According to the Universities Statistics Report for the years 2024 and 2025, a total of 628,541 students were enrolled in tertiary education institutions in the year 2024. This proves that there is a large pipeline of interns across public and private universities.

Internship programmes offered by large well-established organisations are cutthroat, leaving small to medium enterprises which predominantly manage these attachment programs without dedicated HR technology. Intern registration is conducted manually, task delegation is mostly verbal, and logbook entries end up filled just to meet assessment requirements. The certificates of completion end up generic with feedback such as “was a good leader” instead of evidentiary weight.

These conditions were directly experienced during the researcher’s own industrial attachment at a Kenyan organisation operating an enterprise resource planning environment (ERP). Throughout the attachment, there was no formal digital onboarding process, tasks were delegated verbally and tracked through personal notes. The certificate issued at completion (*see appendix*) did not provide achievements or merit-based reports.

It is within this context that an internship management system will bridge the gap between administrative inconvenience and attachment programme credibility.

1.3 Problem Statement

Through the use of paper-based or ad-hoc systems in intern management, organisations forgo benefits they would have gained from a digital alternative such as reduced administrative overhead, operational efficiency, and qualitative capability and talent data.

1.4 Objectives

1.4.1 Main Objective

To design and develop a web-based Intern Management System that digitises task tracking, logbook documentation, and performance reporting for intern attachment programs.

1.4.2 Specific Objectives

- i. To analyse the current challenges facing intern management

- ii. To design the system architecture and role-based mechanisms of the intern management system.
- iii. To develop and implement an integrated web-based intern management system.
- iv. To evaluate the usability of the intern management system against its requirements.

1.5 Research Questions

- i. What are the challenges faced by interns, supervisors, and HR personnel due to the lack of an IMS?
- ii. What entails the system architecture design of the IMS?
- iii. What entails the development of an integrated IMS?
- iv. How does the IMS meet its functional and non-functional requirements?

1.6 Significance of the Study

The development of the IMS is relevant in two interdependent aspects: organisational, and contextual (to interns) based on Kenyan institutional realities.

At the **organisational level**, the IMS offers operational value. By replacing ad-hoc practices with a structured digital system, organisations are able to reduce administrative overhead, improve consistency and traceability of task delegation, and generate evidence-based performance records that support institutional reporting obligations. These benefits are key to organisations managing multiple interns simultaneously across departments.

For **interns**, the system provides a formal documented record of their internship activities including tasks completed, competencies demonstrated, skills gained, and supervisor evaluations. This comprises a substantive professional record compared to the generic certificates of completion that are usually issued. This has direct implications on a graduates employability and perceived value of attachment programs among prospective employers. In addition, the digital logbook management makes their report required in fulfillment of their academia a more efficient and accurate product.

1.7 Scope and Limitations

1.7.1 Scope

The IMS developed in this proposal is scoped to the core operational requirements of a single-intern hosting organisation managing interns through a structured attachment/

intern programme. The system comprises six functional modules: authentication and role-based access control, intern onboarding, task management, digital logbook maintenance, performance evaluation, and automated reporting.

The system supports three primary user roles: Administrator, Supervisor, and Intern, with the additional provision for a guest access role for university supervisors.

1.7.2 Limitations

Several limitations constrain the scope of the study.

First, the primary data collected via structured questionnaire was limited by a low response rate of less than 20 participants. This restricts the statistical generalisability of the survey, hence affecting the choice of methodology selected in Chapter Three.

Second, the system's evaluation is conducted in a controlled academic environment rather than a live organisational deployment. This means the absence of empirical performance metrics such as concurrent user load and full data volumes.

Lastly, the IMS is designed for a Kenyan attachment context.

1.8 Proposed Modules

The IMS is composed of six core modules that comprise an integrated system capable of managing the complete lifecycle from registration through to formal completion reporting.

The **Authentication and Role-Based Access module** forms the system's security foundation, managing the user login, session handling, and the enforcement of role-specific access permissions for Administrators, Supervisors, and Interns. Subsequent modules are dependent on the established access boundaries.

The **Intern Onboarding module** enables administrators to formally register interns within the system, capturing placement details, department assignment, and supervisor allocation.

The **Task Management module** supports the creation, assignment, and tracking of intern tasks. Supervisors assign task with defined parameters including priority and due date. Interns update task statuses and submit deliverables. In turn, supervisors review and approve or return submissions.

The **Digital Logbook module** provides interns with a structured mechanism for documenting their daily and weekly activities. Entries are stored digitally and are accessible to the assigned supervisor. The guest access feature allows external

university supervisors to view a student's logbook via a limited time token URL, without requiring a full system account.

The **Performance Evaluation module** enables supervisors to assess an intern performance against a defined set of competency criteria at the conclusion of the placement. Scores and narrative feedback are recorded formally within the system.

The Reporting and Notifications module generates a formal PDF completion reports aggregating an intern's onboarding, task, logbook, and evaluation data. It also dispatches automated email notifications to relevant users on key system events such as task management, task submission, and logbook updates, reducing the reliance on manual communication.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a review of literature relevant to the design and development of the intern management system (IMS). The review is organised around the four objectives established in Chapter One, each corresponding to a thematic area in the existing consolidated knowledge. The review draws on peer-reviewed journal articles, published papers, and constitutional acts. The chapter concludes with a conceptual map illustrating the key independent and dependent variables identified across literature and their application in the design of the proposed system.

2.2 Review Related To Objective One: To Analyse the Current Challenges Facing Intern Management

The first specific objective of the study is to analyse the current challenges of intern management within Kenyan organisations, mostly small to medium enterprises. The relevant literature prove these challenges are characteristic of broader patterns in human resource management practice.

According to the Employment Act (2007), the definition of employer includes an apprentice or indentured learner. The description of an indentured learner is integrated into that of an intern, where they are mandated to receive a contract of service from their employer. Therefore as per the Employment Act, interns are regarded as employees, though position of “intern” or “attache”, compared to permanent or contractual employment, are usually short-term.

According to the International Labour Office (Employment Working Paper No. 240, 2018), the acquisition of workplace experience skills may occur through traditional formal or informal arrangements, whether or not a specific qualification or certification is issued. This leads to ad-hoc management practices such as informal onboarding, verbal task delegation and the issuance of generic certificates of completion out of mandate as the norm. Informal onboarding is identified as a deficiency that prevents a new employee, in this perspective an intern, from transitioning into an integrated member regardless of their period of service (Meyers & Bartel, 2017).

2.2 Review Related to Objective Two: To design the system architecture and role-based access mechanisms of the Intern Management System

The second specific objective is to design the IMS in accordance with established principles of information systems architecture, database design, security, and user interface engineering.

Role-Based Access Control (RBAC) is a widely adopted access control model in which system permissions are assigned to roles rather than to individual users, and users acquire permissions by being assigned to one or more roles (Ferraiolo & Kuhn, 1992). RBAC is well-suited for information systems such as the IMS, which has clearly defined user categories: the Administrator, Supervisor, and Intern roles. In the Laravel implementation of the IMS, RBAC is enforced through the framework's native Gate and Policy mechanisms, which enable declarative permission definitions that are evaluated at the application layer before any database interaction occurs.

The IMS data model is implemented as a relational database schema in Microsoft SQL Server designed to the third normal form (3NF) ensuring each table represents a single entity. Referential integrity is enforced through foreign key constraints at the database level, complementing the application-layer validation through Laravel Eloquent ORM.

The task management and approval workflow components of the IMS follow the Business Process Management (BPM) theory, which provides conceptual tools for modeling, analysing, and improving activity sequence structures aiding organisations in accomplishing business objectives (Weske, 2019). Wysocki (2019) argues that task management characterised by short task cycles, frequent status transitions, and supervisor review gates, is best served by adaptive workflow models that accommodate task-level iteration rather than requiring upfront specification of all task parameters. This suits the IMS's intern task management, interdependent with the task status model transitioning from Pending, In Progress, Submitted, Approved, Rejected, ensures a structured workflow and triggers notifications at each status change. This provides the intern and supervisor with a shared, persistent and auditable record of task lifecycle hence replacing informal verbal delegation.

The IMS user interface (UI) follows four of Nielsen's (1994) ten usability heuristics. They are *visibility of system status* where interns and supervisors can determine the current state of tasks, logbook entries, and evaluations; *match between system and real world* by using terminology and workflow structures corresponding to intern management practices users are familiar with; *user control and freedom* (by providing clear mechanisms for correcting errors and navigating system functions); and *error prevention* by designing forms and workflows that minimise the possibility of data entry error before they occur. The principle of universal usability (Shneiderman and Plaisant, 2010) is implemented ensuring the system design suites users across a range of technical literacy levels and device types.

2.3 Review Related to Objective Three: To Develop and Implement an Integrated Web-based Intern Management System Application

The third specific objective of the study is the development and implementation of the web-based IMS.

The chosen software development methodology is the Agile methodology. Agile software development represents a family of iterative, incremental development approaches that prioritise working software, adaptive planning, and continuous stakeholder collaboration over comprehensive upfront documentation and rigid process adherence (Beck et al., 2001). It suites the study's context: a single-developer environment enabling incremental iteration validation against stakeholder feedback.

The PHP/ Laravel framework was selected as the primary development technology for the IMS. Laravel is a full-stack MVC web application framework that provides built-in support for authentication, routing, Eloquent ORM, queue management, and PDF generation (Otwell, 2023). Microsoft SQL Server (MSSQL) was selected as the database management with T-SQL as the query language. These decisions directly support the IMS's functional requirements as documented in Chapter Three.

2.4 Review Related to Objective Four: To Evaluate the Usability of the Intern Management System Against its Requirements

The fourth specific objective of the study is to test and validate the IMS against its stated functional and non-functional requirements. The review delves into the execution of the objective through three activities: software testing, user acceptance testing (UAT), and non-functional requirements testing.

Myers, Sandler, and Badgett (2011) define software testing as the process of executing a program with the intent of finding errors, and distinguish it between two fundamental testing orientations: black-box testing and white-box testing. Black-box testing evaluates system behaviour from the perspective of the user without knowledge of internal implication details while white-box testing evaluates the internal logic, data flows, and control paths of the code itself. Black-box testing is the primary method for validating functional requirements by simulating the perspective of the key users (Administrator, Supervisor, Intern) hence confirming the system behaves as in the use cases and user stories in Chapter Three. White-box testing is applied to selectively to critical components, including the RBAC permission layer, the guest token generation, and validation mechanism as per user workflows.

Pressman and Maxim (2020) emphasise that non-functional requirements testing is critical to system quality assurance as it reduces the chances of failures such as browser incompatibilities and security vulnerabilities. Security testing evaluates the RBAC implementation, password hashing, and token-based guest access mechanism. Compatibility testing confirms that the system is accessible and functional across the browser environments specified in NFR 07 in Chapter Three. Lastly, performance testing assesses the system's response time of three seconds under simulated concurrent user load and individual users under typical usage conditions as per NFR 02 in Chapter Three.

User Acceptance Testing (UAT) is the final phase of the software testing process, in which the system is evaluated by representative users in conditions that simulate the intended operational environment (Satzinger et al., 2016). For the IMS, UAT is conducted by presenting the sample MVP to a representative sample of users (internn and supervisors) and evaluating their interaction with the system against the user

stories documented in Chapter Three. The findings are used to identify usability deficiencies and functional gaps unidentified during developer-conducted testing.

2.6 Concept Map/ Conceptual Map

The conceptual map presented in figure 1 illustrates the relationship between key constructs and the proposed IMS. The map is organised around three layers: the independent variables (the theoretical frameworks that inform the system's design), the mediating construct (the system itself), and the dependent variables (the organisational outcomes the system is designed to produce).

The key concepts comprise six theoretical constructs drawn from literature:

Agile methodology: An iterative and incremental framework that emphasises adaptive planning, early and continuous delivery of working software, and close collaboration between the development team and stakeholders (Beck et al., 2001). It is the chosen development methodology for the IMS as documented in Chapter Three.

Model-View-Controller architecture: It guides the design of the IMS by decoupling the webapp functionality from the informational content through the use of the model (contains application-specific content and processing logic), view (contains interface specific instructions), and controller (manages the flow of data to between the model and controller) (Pressman and Maxim, 2020).

Role-Based Access Control theory: It guides the workflow design of the IMS, specifically task management and approval workflow design (Wysocki, 2019).

e-Portfolio Theory: It guides the

Business Process Management (BPM) and workflow theory:

Usability heuristics:

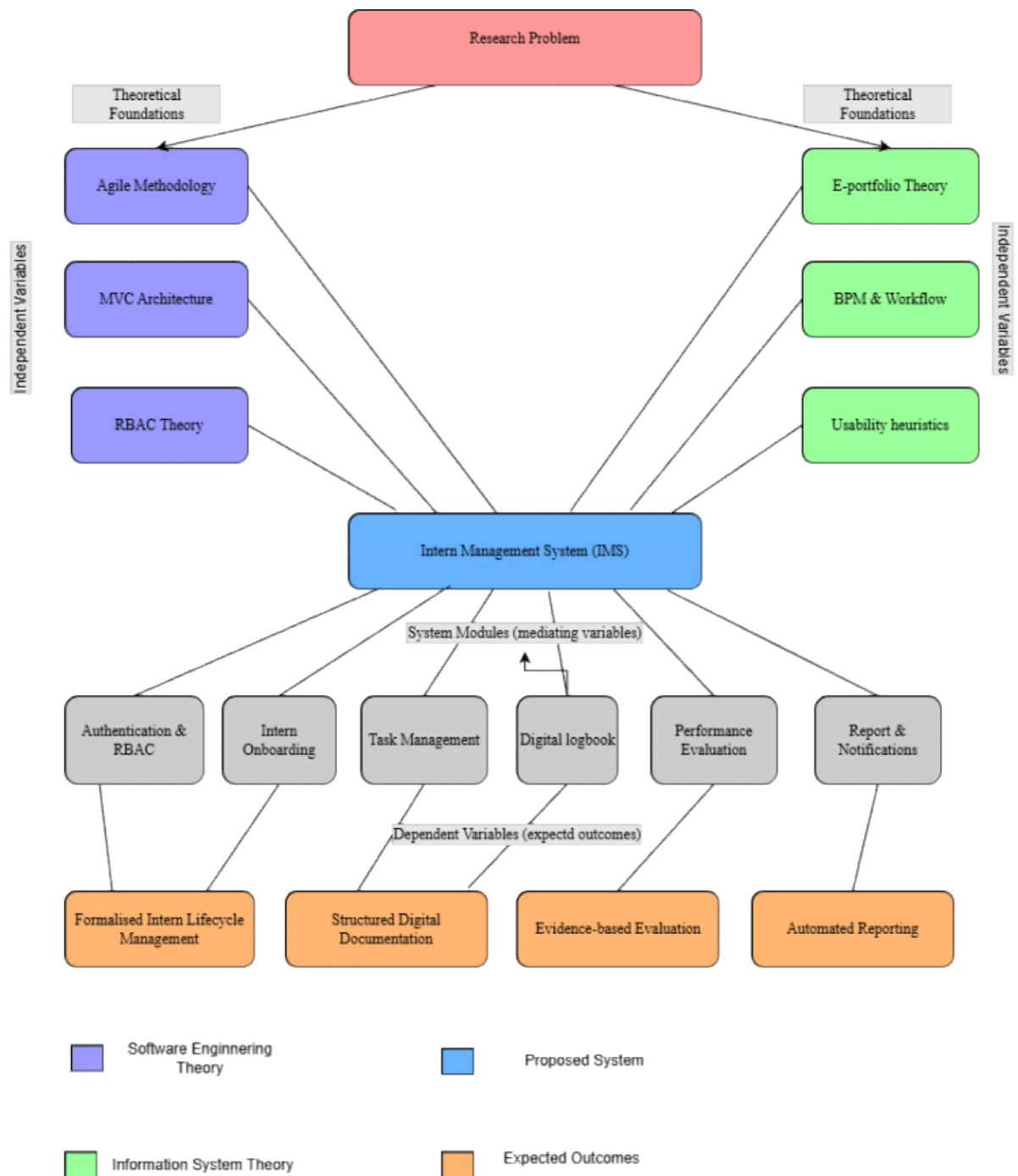


Figure 1: Conceptual Map of the IMS

CHAPTER THREE

METHODOLOGY

3.1 Introduction

The chapter begins by identifying the research and development methodology adopted for the design and implementation of the Intern Management System (IMS). It proceeds to give a comprehensive analysis of system requirements, followed by the design decisions made to transform those requirements into an implementable architecture. The chapter draws on data gathered through direct observation and document review, including internship artefacts such as certificates of completion to establish both functional and non-functional requirements for the system.

3.2 Research Methodology

The research design adopted for the system is a qualitatively dominant mixed-methods approach, a converge parallel design in which qualitative and quantitative data were collected concurrently and compared to assess the consistency of their findings (Creswell & Creswell, 2018).

The qualitative strand of this study comprises two complementary methods: **direct observation** of intern management practices conducted during the researcher's own industrial attachment at a Kenyan organisation, and **documentary analysis** of intern management artefacts including certificates of completion, manual logbook templates, and institutional onboarding documentation. These qualitative sources offer direct, unmediated access to the operational conditions the IMS is designed to address. The quantitative strand comprised a structured questionnaire distributed to current and former interns and organisational supervisors, designed to assess the prevalence of common intern management deficiencies including the absence of formal onboarding, unclear task delegation, and reliance on paper-based logbooks. Fourteen responses were received.

While this sample is insufficient to support statistical generalisation, the questionnaire data is treated as corroborative of the qualitative findings rather than as an independent evidential basis.

The three data sources were subjected to **concurrent data triangulation**, a strategy in which independently gathered evidence from multiple sources is compared at the analysis stage to assess convergence (Denzin, 1978; Creswell & Creswell, 2018). The questionnaire captures the perspectives of individuals beyond the researcher's own experience, but is subject to social desirability bias and was constrained in this study by a limited response rate. Direct observation provides unfiltered, insider access to actual practice rather than reported practice, but is bounded by a single organisational site and a single attachment period. Document analysis yields evidence of processes reflecting current practices. Despite these limitations, all three sources converge on a consistent finding: intern management in the Kenyan organisational context is characterised by informality, absent digital infrastructure, and inadequate documentation.

3.2.1 Development Methodology

Development Approach

The Agile software development methodology was selected as the development approach for the IMS. Agile is an iterative framework that emphasises adaptive planning, early and continuous delivery of working software, and close collaboration between the development team and stakeholders (Beck et al., 2001). This methodology is suitable to the project for several reasons.

First, the IMS is a moderately scoped academic project with a single developer. This makes the overhead of process frameworks such as the Waterfall model or the DevOps methodologies disproportionate to the project's scale. DevOps being suitable to multi-team production environments with continuous integration and deployment does not apply to a single-developer academic build while MLOps being designed for machine learning is outside the scope of the IMS.

Second, Agile emphasises on producing working software at the end of each iteration. This aligns with the expected outcome of this project: a functional MVP with six core modules, hence making it the ideal development methodology.

Iterative Process

The development of the IMS is divided into six discrete iterations, each corresponding to a core system module. This approach ensures that each component is fully implemented and tested before the next build, reducing integration risk and maintain a coherent and functional system at each phase.

Each phase is assigned a time frame of 1-2 weeks involving the following stages:

Planning: Involves defining the iteration scope and its specific features

Analysis and Design: Entails detailing the requirements for the selected features

Implementation: Building the iteration's module based on the design

Testing: Evaluating if the module performs as per its designated function

Review and Feedback: Demonstrating the completed features to stakeholders and using feedback to ensure it meets their requirements.

<i>Iteration</i>	<i>Module</i>	<i>Key Deliverables</i>	<i>Rationale</i>
<i>Iteration 1</i>	Authentication and RBAC	Define clear user roles: Administrator, Supervisor, Intern	Sets access boundaries, data visibility rules, and approval workflows of all modules
<i>Iteration 2</i>	Intern Onboarding	Enable administrator to perform intern registration and departmental assignment.	Create onboarding workflows for said events.
<i>Iteration 3</i>	Task Management	Enable assignment creation, task delegation, and status tracking	Create supervisor approval workflows
<i>Iteration 4</i>	Digital Logbook	Provide interns with a structured recording mechanism for entries of daily and weekly activities	Provide guest supervisor access to view logs using token-based authentication
<i>Iteration 5</i>	Performance Evaluation	Develop the performance evaluation and merit reporting module	Create a merit-based system based on the severity of the accomplished tasks.
<i>Iteration 6</i>	Reporting and Notification Layer	Develop automated notifications, and link it to SSRS for report generation	Generate formal downloadable PDFs for requested reports.

Table 1: Summary of module iterations

3.3 Data Collection Methods

3.3.1 Questionnaire Survey

A structured questionnaire was designed and administered digitally via Google Forms to a purposive sample of individuals with direct experience of intern management processes, comprising current and former interns and organisational supervisors operating within Kenya. The instrument comprised closed-ended items addressing the availability of formal onboarding processes, the predominant mode of task assignment and tracking, the format and accessibility of intern logbooks, the frequency of structured supervisor feedback, and the overall adequacy of existing intern management practices. Closed-ended items were selected to facilitate numerical coding and to enable comparability across respondents.

Fourteen responses were received within the two-week distribution period. This response rate, while limited, does not invalidate the instrument's contribution to the study. As noted in Section 3.2, the questionnaire functions within a triangulation strategy as a corroborative source rather than the primary evidential strand. Where questionnaire findings align with observational and documentary evidence — as they do on the central finding of pervasive informality and absent digital infrastructure — they strengthen the overall evidentiary base. Where they diverge, that divergence is noted and interpreted rather than suppressed. This approach to limited quantitative data is consistent with the convergent parallel design adopted for this study and with established practice in mixed-methods information systems research (Venkatesh et al., 2013).

3.3.2 Direct Observation

Direct observation was conducted during the researcher's own industrial attachment at a Kenyan organisation operating an ERP environment. Observation data was recorded systematically, noting the absence of a digitised formal onboarding process, automated reporting, and the reliance on verbal task delegation to interns despite the being regarded as, albeit temporary, a part of the organisation. This first-hand observation constitutes sample insider knowledge of the operational realities in SMEs within the country. In addition, it included the analysis of other enterprise-popular ERP systems such as Sage and Zoho, which also do not account for intern management.

3.3.3 Document Analysis

Document analysis was conducted on a sample intern management artefacts, including a certificate of completion issued at the end of the researcher's industrial attachment, and sample paper-based logbook page. These documents were examined for the information they captured, the processes they implied, and the gaps they revealed. The certificate of completion, for example, was found to only contain the intern's name, the programme's duration, and a generic statement of completion lacking references to their specific competencies. This finding directly informs the IMS merit reporting module, which is designed to generate structured, evident-based completion reports, a better alternative to the generic certificates currently in use.

3.4 System Analysis and Design (SAD)

3.4.1 System Analysis

Requirements Gathering

Questionnaire Survey: A structured questionnaire was designed and published digitally via Google Forms with the goal of collecting substantive quantitative data. The response though brief (14 in a span of a month), does not invalidate its relevance but contributes to the requirements.

Direct Observation: Direct observation was conducted during the researcher's industrial attachment at a Kenyan organisation operating an ERP. The data included the absence of a digital onboarding process, reliance on verbal and informal task delegation, and infrequent supervisor check-ins.

Document Analysis: Document analysis was conducted on sample intern management artefacts such as manual logbook page templates and certificates of completion.

Requirements Analysis

Requirements analysis involves refining raw data gathered during requirements gathering into formal structured specifications that are unambiguous, complete, and testable. (Pressman & Maxim, 2020).

Functional Requirements

The functional requirements define what the IMS must do. This comprises the specific behaviours and capabilities the system must provide to its users. the following were identified for the MVP:

FR-01: The system shall support three distinct users: *administrator*, *supervisor*, and *intern*, with role-specific access permission enforced at the application layer.

FR-02: The system shall allow administrators to register new interns, capturing personal details, departmental assignment, placement dates, and assigned supervisor.

FR-03: The system shall enable supervisors and administrators to create and assign tasks to interns, specifying task title, description, priority, due date, and expected deliverable.

FR-04: The system shall enable supervisors and administrators to create and assign tasks to interns, specifying task title, description, priority, due date, and expected deliverable.

FR-05: The system shall allow interns to update the status of assigned tasks and submit deliverables for supervisor review and approval.

FR-06: The system shall provide a digital logbook module for dated daily and weekly activity entries, including tasks performed, challenges encountered, and skills developed.

FR-07: The system shall generate a secure, time-limited guest access token enabling external university supervisors to view an intern's logbook without requiring a full system account.

FR-08: The system shall enable supervisors to evaluate intern performance against predefined competency criteria and record structured feedback.

FR-09: The system shall generate a formal PDF report summarising an intern's onboarding completion, tasks assigned and completed, logbook entries, and performance evaluation scores at the conclusion of an internship.

FR-10: The system shall send automated email notifications to relevant users upon key system events, including task assignment, task status updates, and logbook submission.

Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints that govern how the IMS performs, rather than what it does. The following non-functional requirements were identified:

NFR-01 (Security): The system shall implement authentication using hashed passwords and shall enforce role-based access control to prevent unauthorised access to data of functionality.

NFR-02 (Performance): The system shall return page responses within three seconds for individual authenticated users under typical usage conditions. Concurrent load handling is addressed through architectural design decisions documented in section 3.4.2.

NFR-03 (Usability): The system's user interface shall be intuitive and navigable by users with basic computer literacy, requiring no specialised technical training.

NFR-04 (Reliability): The system shall be available during normal working hours (8:00 AM to 6:00 PM EAT, Monday to Friday) with a target uptime of 99%.

NFR-05 (Maintainability): The system codebase shall be structured according to the Model-View-Controller (MVC) design pattern as enforced by the Laravel framework, ensuring modularity and ease of future maintenance.

NFR-06 (Scalability): The system architecture shall be designed to accommodate growth in the number of registered interns and organisations without requiring significant structural modification.

NFR-07 (Compatibility): The system shall be accessible via modern web browsers (Chrome, Firefox, Edge, Safari) on both desktop and mobile devices without loss of core functionality.

NFR-08 (Data Integrity): The system shall use database-level constraints, including foreign keys and not-null constraints, to maintain referential integrity across all related entities.

Use Cases

Use cases describe the interactions between the system and its users (actors) in relation to the goals the actors seek to accomplish. The primary actors of the IMS are the administrator, supervisor (internal), university supervisor (external/ guest), and the intern. Key use cases include:

UC-01: Register Intern — The Administrator registers a new intern by entering required personal and placement details. The system creates a user account with the Intern role and initiates the onboarding checklist.

UC-02: Assign Task — The Supervisor creates a new task, specifies its parameters, and assigns it to one or more interns. The system notifies the assigned intern(s) by email.

UC-03: Update Task Status — The Intern logs into the system, navigates to their task list, and updates the status of an assigned task. Upon marking a task as complete, the Supervisor receives a notification for review.

UC-04: Submit Logbook Entry — The Intern creates a new logbook entry, recording activities for a given period. The entry is saved and becomes visible to the assigned Supervisor.

UC-05: View Logbook (Guest) — The University Supervisor accesses the system via a shared token URL. The system validates the token and presents the relevant intern's logbook in read-only mode.

UC-06: Evaluate Intern — The Supervisor completes a structured performance evaluation form for an intern, scoring competency areas and providing narrative feedback. Scores are saved to the intern's record.

UC-07: Generate Report — The Administrator selects an intern and requests a completion report. The system compiles the intern's onboarding, task, logbook, and evaluation data and produces a formatted PDF.

User Stories

User stories are short informal descriptions of a feature from the perspective of an end user, in the format: "As a [role], I want to [action], so that [benefit]" (Cohn, 2004). Below, are user stories capturing the core functionalities of the IMS:

US-01: As an Administrator, I want to register new interns and assign them to supervisors, so that all placements are formally recorded in the system from day one.

US-02: As an Intern, I want to view all tasks assigned to me in a single dashboard, so that I can prioritise my work and track my progress against deadlines.

US-03: As a Supervisor, I want to receive automatic notifications when an intern marks a task as complete, so that I can review and approve deliverables promptly.

US-04: As an Intern, I want to write and save digital logbook entries for each working week, so that my activities are documented without reliance on a physical logbook.

US-05: As a University Supervisor, I want to access a student's digital logbook via a shared link without needing a system account, so that I can fulfil my monitoring obligations conveniently.

US-06: As a Supervisor, I want to complete a structured evaluation form at the end of an internship, so that my assessment is recorded formally and consistently across all interns.

US-07: As an Administrator, I want to generate a PDF completion report for any intern at the end of their placement, so that I can provide them with a detailed, evidence-based record of their internship.

Feasibility Study

Technical Feasibility

Technical feasibility is about whether the technologies required to build and operate the proposed system are available, mature, and within the competencies of a single-developer environment.

The IMS is built on the PHP/ Laravel framework and Microsoft SQL Server (MSSQL). Laravel provides a robust MVC framework with built-in support for authentication, routing, Eloquent ORM, and PDF generation via packages such as DomPDF, which directly support the IMS's functional requirements. MSSQL provides the relational database capabilities required to manage the complex entity relationships of the IMS data model, using T-SQL as the scripting language. The front-end will be implemented using standard web technologies (HTML5, CSS3, Bootstrap, and Laravel Blade templating), all which are free and browser-compatible. Therefore, the IMS is concluded to be technically feasible.

Economic Feasibility

Economic feasibility assesses whether the costs associated with developing and operating the system are justifiable relative to the expected benefits. The development cost of the IMS is primarily measured in terms of developer time, as the software stack, PHP, Laravel, MSSQL Express, and the associated open-source libraries, is available at no monetary cost for development purposes. Deployment costs are expected to be minimal for the academic MVP, as the system can be hosted on a local server or a low-cost cloud instance with the domain cost and SSL certificates approximating KES 1,500 as per TrueHost pricing. Therefore, the IMS is concluded to be economically feasible.

Operational Feasibility

Operational feasibility assesses whether the proposed system will be accepted and used effectively within the target organisational environment.

The primary operational risk for the IMS is supervisor adoption. Supervisors who currently rely on informal, verbal intern management practices may resist the additional administrative demand of logging into a system to assign tasks, review logbooks, and complete structured evaluations. This risk is mitigated by the IMS's

interface design which prioritises simplicity, speed, and efficiency by transforming the overhead to an intern's responsibility. This frames interns as a talent contingency workforce rather than administrative burden. Moreover, automated email notifications reduce the frequency with which supervisors must actively log into the system.

Therefore, the IMS is concluded to be operationally feasible, subject to appropriate change relative to stakeholders' needs.

System Modelling

Data Flow Diagram

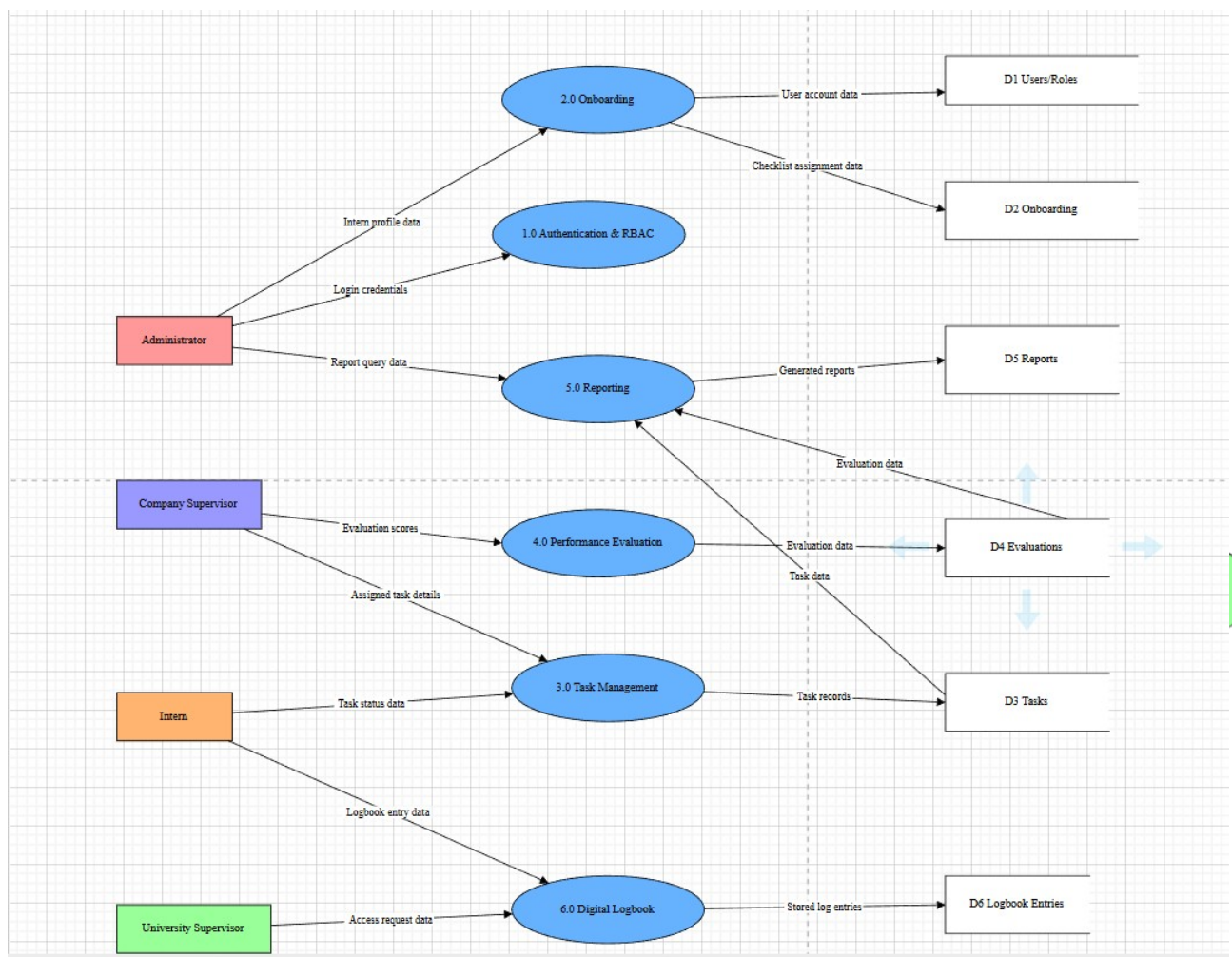


Figure 2: Level 1 DFD using Yourdon DeMarco's notation

Entity Relationship Diagram (ERD)

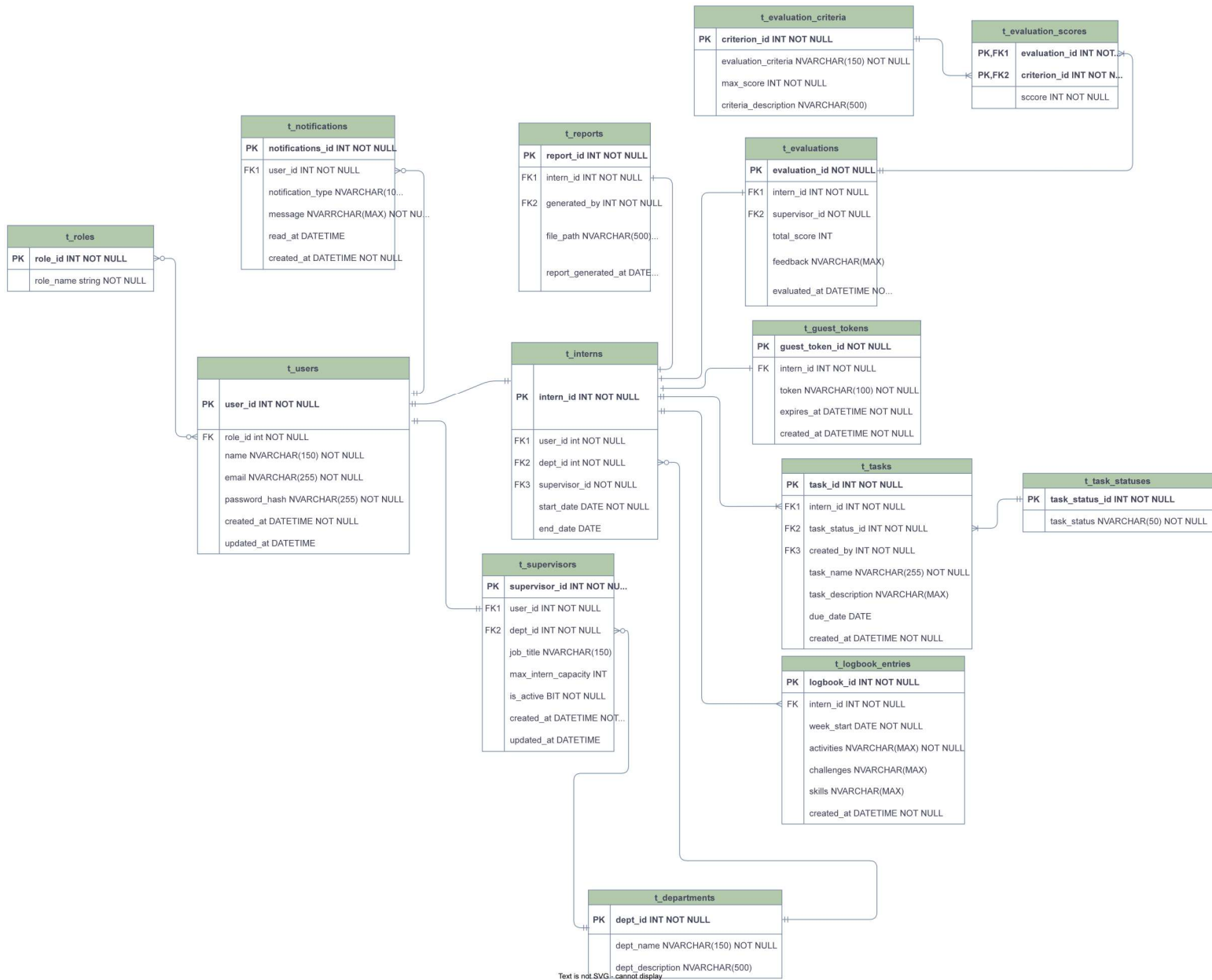


Figure 3: ERD of the IMS using table notation

Unified Modelling Language Diagram (UML)

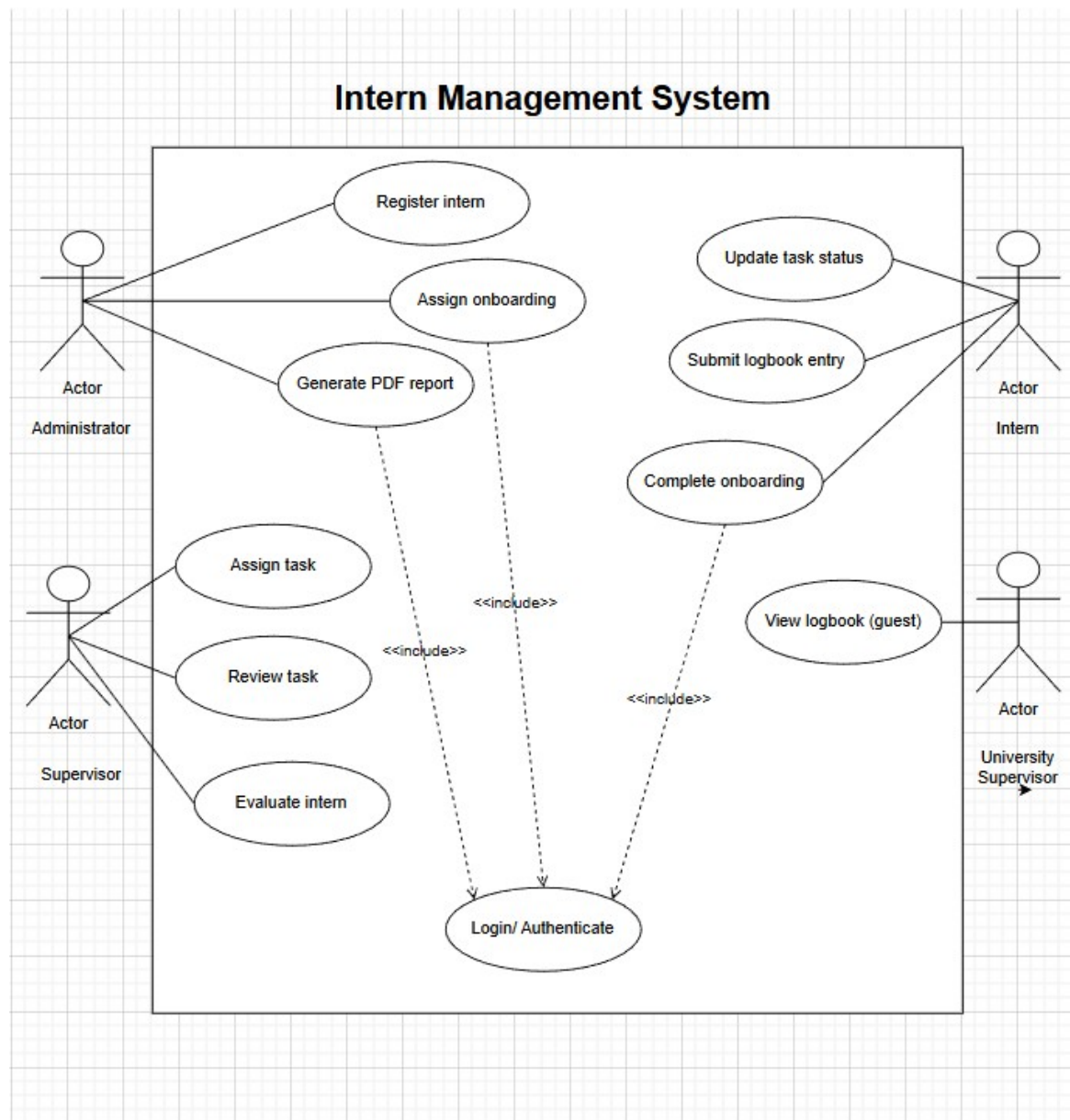


Figure 4: UML diagram of the IMS

3.4.2 System Design

Architectural Design

System Architecture

The IMS is implemented as a monolithic web application following a client-server architecture. The client layer comprises standard web browsers accessing the application via HTTP/HTTPS. The server layer hosts the Laravel PHP application and the Microsoft SQL Server database instance. For the MVP, both the application and

database servers may reside on a single host. In a production deployment, these would be separated for performance and security reasons.

Hardware components include a web server (minimum 2 vCPUs, 4 GB RAM for development; higher specifications recommended for production), a database server running SQL Server 2019 or later, and client devices (any device with a standards-compliant web browser). Network components include an Ethernet or Wi-Fi local area network for development and a secure HTTPS connection over the public internet for production deployment. Software components include Ubuntu Server 22.04 LTS or Windows Server 2019 as the operating system, PHP 8.2, the Laravel 11 framework, MSSQL Server 2019, and NGINX or Apache as the web server process.

Layered Architecture

The IMS adopts a three-layer architecture consistent with the MVC pattern enforced by Laravel:

Presentation Layer: This layer encompasses the user interface and is implemented using Laravel Blade templates rendered on the server and delivered as HTML to the client browser. Bootstrap 5 provides the responsive CSS framework. This layer handles user input, displays data retrieved from the business logic layer, and enforces front-end validation rules.

Business Logic Layer: This layer is implemented in Laravel Controllers and Service classes. It processes user requests, enforces business rules (such as role-based access control and task approval workflows), orchestrates interactions between the presentation and data access layers, and triggers notifications and report generation.

Data Access Layer: This layer is implemented using Laravel's Eloquent ORM, which provides an abstraction over raw SQL queries and maps PHP model objects to database tables. Eloquent models define entity relationships and encapsulate data access logic, ensuring that database interaction is centralised and consistent.

Detailed Design

Component Design

The IMS is composed of six principal software components, each corresponding to a core functional module, each implemented as a set of related Laravel Controllers, Models, Views, and Service classes. The Authentication and RBAC component

manages user session handling, password hashing, and permission enforcement using Laravel's built-in Gate and Policy mechanisms. The Onboarding component manages intern registration workflows and checklist assignment. The Task Management component handles task CRUD operations, status transitions, and approval notifications. The Digital Logbook component manages entry creation, retrieval, and guest access token generation and validation. The Evaluation component manages performance scoring and feedback storage. The Reporting and Notifications component generates PDF reports using DomPDF and dispatches email notifications using Laravel's Mail facade and queue system.

Database Design

The physical database schema is implemented in Microsoft SQL Server and is derived directly from the ERD presented in Figure 3.4. The primary tables in the IMS schema are listed below, with key columns and relationships described:

users: Stores all system user accounts (id, name, email, password_hash, role_id, created_at, updated_at).

roles: Defines the three system roles (id, name: Administrator | Supervisor | Intern).

interns: Stores intern-specific profile data (id, user_id [FK → users], department_id [FK → departments], supervisor_id [FK → users], start_date, end_date).

User Interface Design

The IMS user interface is designed according to the principles of simplicity, role-appropriate information display, and accessibility. Nielsen's four out of ten usability heuristics (Nielsen, 1994) serve as the design reference framework. These include: visibility of system status, match between system and the real world, user control and freedom, and error prevention. The interface is fully responsive, adapting to both desktop and mobile viewports using Bootstrap 5's grid system.

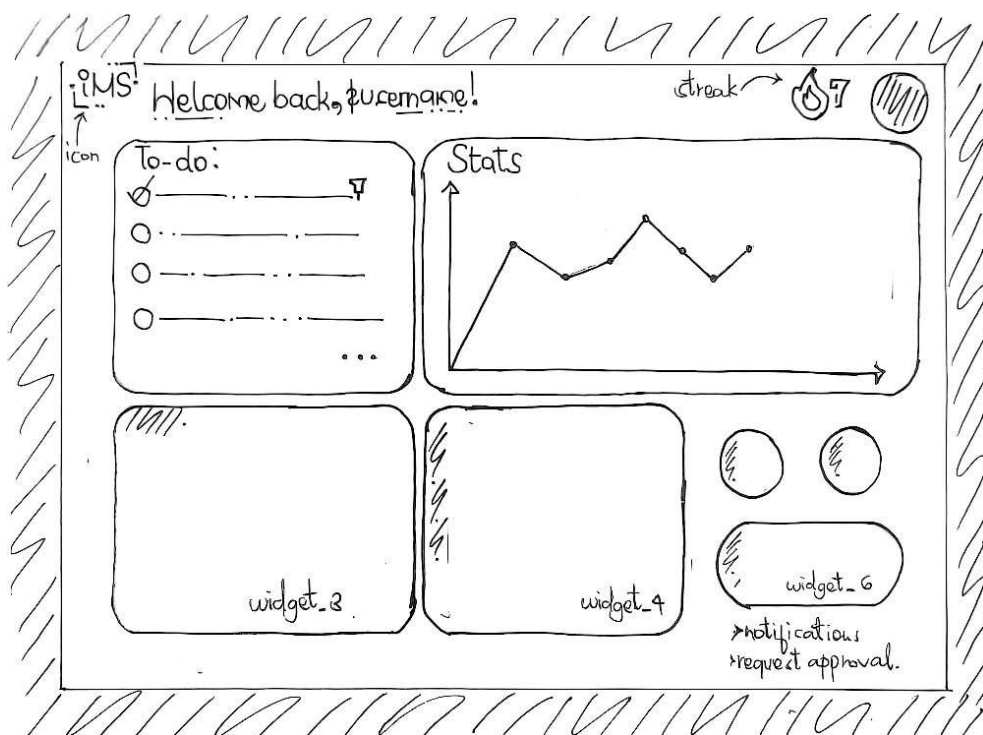
Role-specific dashboards are a core design decision. The Administrator dashboard surfaces intern registration statistics, onboarding progress, and pending report requests. The Supervisor dashboard highlights tasks requiring review and interns due for evaluation. The Intern dashboard displays pending tasks, logbook entry reminders, and onboarding checklist completion status. This role-tailored information

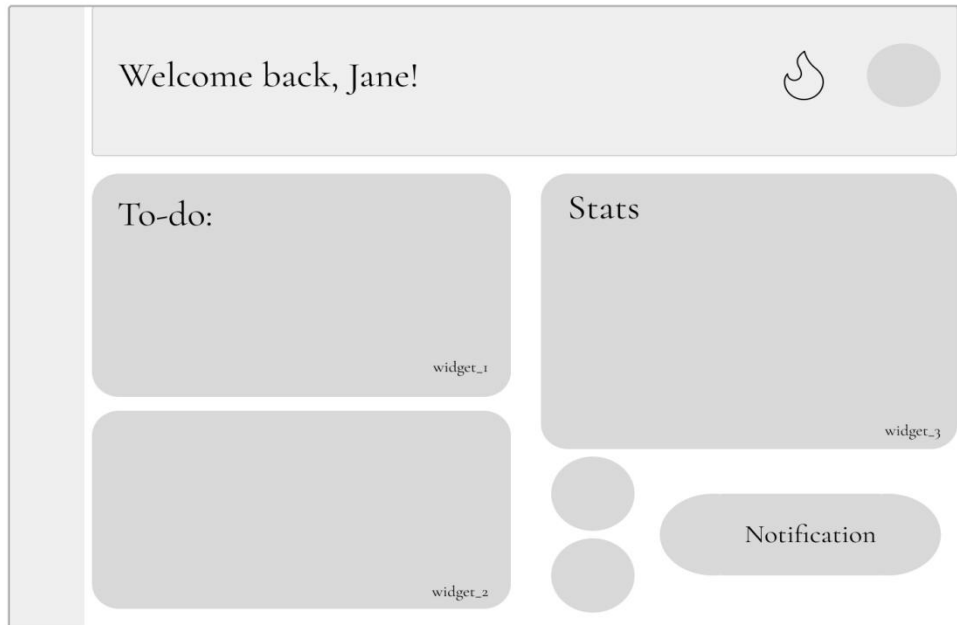
architecture reduces cognitive load by presenting only the data most relevant to each user's responsibilities.

Prototyping

Low-Fidelity Prototype

Low-fidelity prototyping was the initial design step for the IMS user interface, using hand-drawn sketches. Wireframes were produced for the intern dashboard and sidebar menu. Low fidelity prototypes were later implemented with inspiration from the hand-drawn wireframes.





High-Fidelity Prototype

High-fidelity prototypes were developed to provide a realistic representation of the system's final visual appearance and interaction behaviour. These prototypes incorporate the actual colour scheme, typography, form controls, navigation components, and data visualisation elements that will be used in the implemented system. The base design below was used to develop the dashboard.

Design Specification

Technical specifications

The following are the technical specifications summarise the hardware and software, and infrastructure requirements for the development and deployment of the IMS:

<i>Component</i>	<i>Development Specification</i>
Operating System	Windows 10+ or Ubuntu 20.04 LTS+
Programming language	PHP 8.2+
Framework	Laravel 11+
Database server	MSSQL 19+
Web Server	Apache or NGINX

Browser Support	Chrome, Brave, Firefox, Safari
RAM (production)	$\geq$ 4GB
Network components	Ethernet or Wi-Fi

Design Documentation

The design documentation for the IMS comprises all diagrams, descriptions, and specifications presented previously in the chapter. It includes the UML use case diagram which models actor-system interactions, the ERD which documents the data model, the DFD which documents the data flow across system processes, the wireframes and low-fidelity prototypes which document the user interface design.

Review and Validation

Design Review

The IMS design review was conducted in two forms. First, an internal self-review was performed by the researcher, cross-referencing each functional requirement (FR-01 through FR-10) against the design documentation to confirm that every requirement is addressed by at least one identified system component, database entity, or interface screen. Second, a structured review was conducted with the project supervisor (academic lecturer), during which the system architecture, ERD, and wireframes were presented and feedback was incorporated into the design before finalisation. This iterative review process is consistent with the Agile development philosophy adopted

Validation

Validation for the IMS was conducted by mapping each user story (US-01 through US-07) against the design components — use cases, interface screens, database entities, and system processes — to verify that each story's implied functionality is fully supported by the design. Additionally, the high-fidelity prototypes were walkthrough-tested against the identified use cases to confirm that all user flows are navigable without dead ends or missing interface elements. Any gaps identified during validation were resolved by updating the relevant design artefact before the design was declared complete. This pre-implementation validation reduces the

probability of requirement defects propagating into the implementation phase, where they are more costly to correct.

3.5 Research Ethics

Intellectual Honesty: All resources used in the study are referenced and cited to acknowledge authors fully for their work.

Informed Consent: System testing will be done through voluntary participation. Sample documents used in the proposal are gained through consent with sensitive information censored out.

Data Privacy and Security: The application will implement RBAC, password hashing, secure token-based guest access, and targeted security testing to maintain data integrity.

Intellectual Property: Transparency of the application's intellectual property will be maintained through appropriate licencing and compliance to Third-party resources including implemented Laravel libraries.

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APPENDIX

29th December 2025

Dear Sir/Madam,

TO WHOM IT MAY CONCERN

RE: AURELIA WARUGURU MUNDIA ID NUMBER: 42547770

is one of the leading developers of IT Solutions for the financial sector in Kenya and globally.

We are pleased to inform you that Aurelia Waruguru Mundia has completed her internship opportunity at the working for a period from 8th September 2025 to 29th December 2025. She is a well-organized individual who displayed great leadership qualities and worked with a positive attitude towards the tasks assigned to her. She also worked well with various members of the organization and proved high levels of commitment.

We wish her all the best in her future endeavors.

Kind regards,



HEAD OF HUMAN RESOURCES

for

ADDRESS

Nairobi, Kenya

TELEPHONE

WEB

WEEK ONE

Dates: From 12/01/2026

To 16/01/2026.

PART I - To be filled by the Attaché

DAY	NOTES ON WORK DONE
Monday	Reported to the office and was received by the supervisor. I was introduced to the working environment and briefed on KRA operations. I familiarized myself with the office setup, ICT equipment. I observed how staff and interns handle & perform their duties.
Tuesday	I assisted and observed interns during asset movement from Eldoret to Kapenguria KRA offices. I observed ICT equipment being verified, recorded and issued. I learned that ICT assets are tracked and registered to ensure safety, proper usage and accountability.
Wednesday	I observed and assisted interns in troubleshooting a printer that was not printing and was displaying error codes. The troubleshooting process involved checking printer's network connection, printer status and print queue. I was also trained by an intern on crimping ethernet cable by stripping the cable, arranging wires in correct order and attaching an RJ45 connector. The cable I crimped was tested and confirmed working.
Thursday	I was shown how to upgrade from Office 2016 to Office 2024 using organization's server. Also accompanied ICT staff to Kenya Pipeline Company and Eldoret International Airport, where I observed routine maintenance of network devices including routers and switches. Also observed verification and activation of network ports.
Friday	I observed a laptop and printer connection. This involved manually adding the printer, verifying network connectivity, checking printer status and confirming correct printer driver was installed. A test print was performed to confirm successful communication between the laptop and the printer. Also assisted in receiving ICT support calls, where users reported system and hardware issues for further troubleshooting and resolution.